Machine Learning Course Report

Logistic Regression, Feature Selection, PCA, and Dimensionality Reduction

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Course: Data Analysis

Introduction

This report explains the work completed in both practical sessions and includes the graphs used to support the analysis. In the first notebook, I studied a colon cancer dataset with logistic regression and coefficient-based feature selection. In the second notebook, I used PCA first as an exploratory method on the decathlon dataset and then as a dimensionality-reduction method before classification on the digits dataset. I tried to keep the report clear and direct by focusing on what I actually did, what each graph shows, and what I concluded from it.

Lab 1 - Logistic Regression and Feature Selection

In Lab 1, I worked with a binary classification problem where the goal was to predict whether a tissue sample was normal or tumoral from gene-expression values. I first checked the dataset structure, then explored the variables visually, trained logistic regression on all features, extracted the most important coefficients, and finally retrained the model on smaller feature subsets to compare performance.

Feature	Coefficient	Absolute importance
RNF43	0.928	0.928
SLC7A5	0.892	0.892
UGP2	-0.707	0.707
DAO	-0.695	0.695
NEURL1B	-0.627	0.627
NCAPH	0.594	0.594
DDIT4	0.500	0.500
CLIC5	-0.447	0.447
NIBAN1	-0.444	0.444
DHRS11	-0.444	0.444

Lab 1 Model Comparison

To make the comparison clearer, I added a table that summarizes the logistic-regression versions used in the first lab. This helps show directly how the scores change when I reduce the number of genes.

Model	Features used	Accuracy	AUC
All features	60	1.000	1.000
Top 3	3	0.981	0.999
Top 5	5	1.000	1.000
Top 10	10	1.000	1.000

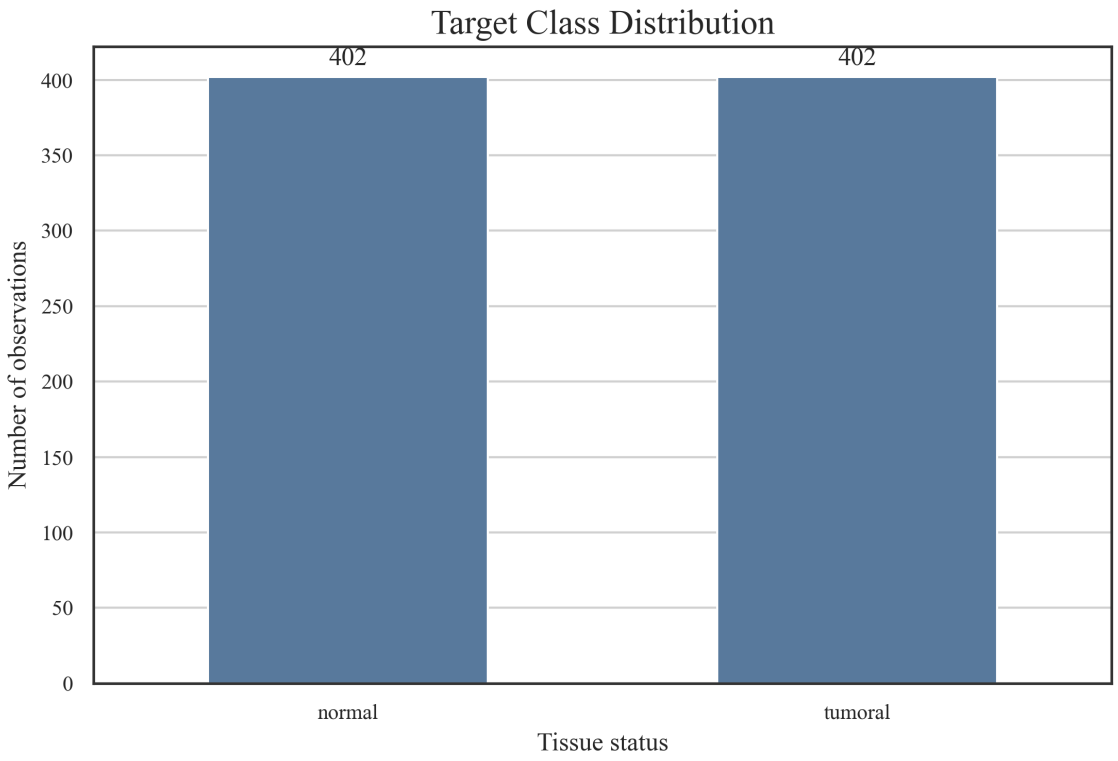


Figure 1. Distribution of normal and tumoral samples.

I started with the target distribution because it tells me whether the classification problem is balanced. Here the two classes have the same number of observations, so the model is not being helped by a dominant class. That makes the later accuracy results easier to trust.

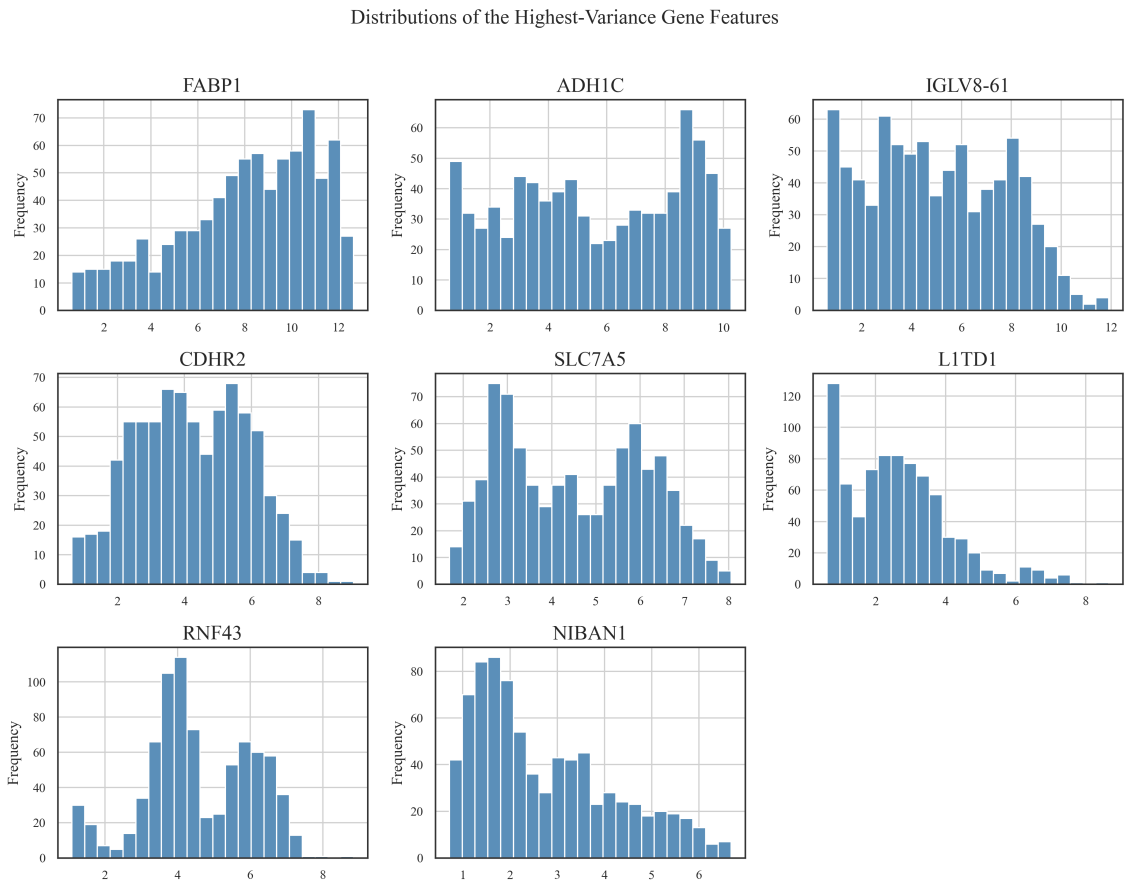


Figure 2. Histograms of high-variance gene-expression variables.

These histograms gave me a first overview of how spread out the main gene-expression values are. I focused on the highest-variance variables because they usually carry more visible signal. This step helped me see that the dataset has real variation and is not just a flat matrix of similar values.

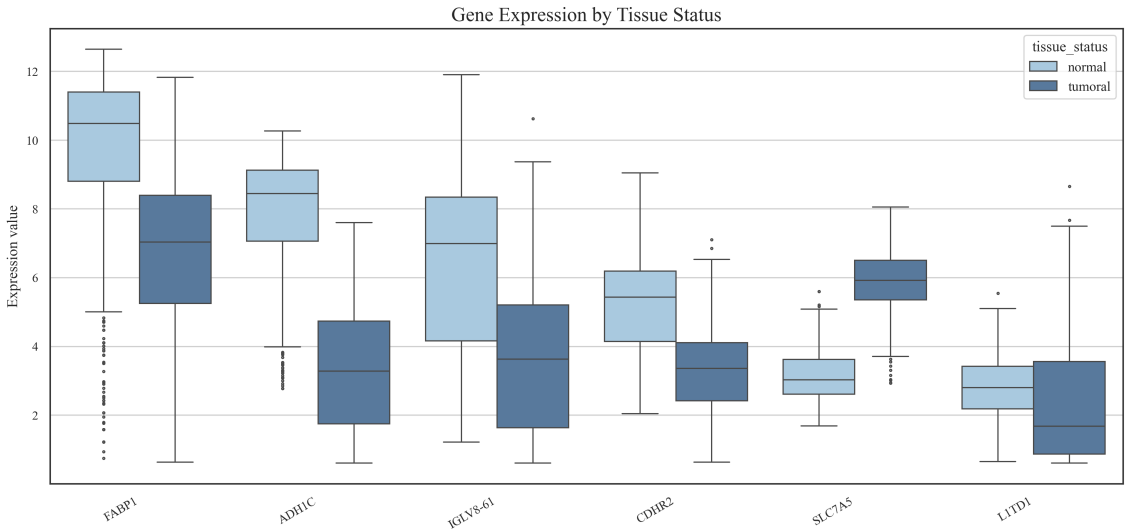


Figure 3. Boxplots comparing selected genes across the two classes.

This figure is useful because it shows class separation directly. Several genes have noticeably different centers and spreads between normal and tumoral tissue. Before even fitting the logistic model, I could already see that some variables were likely to become important predictors.

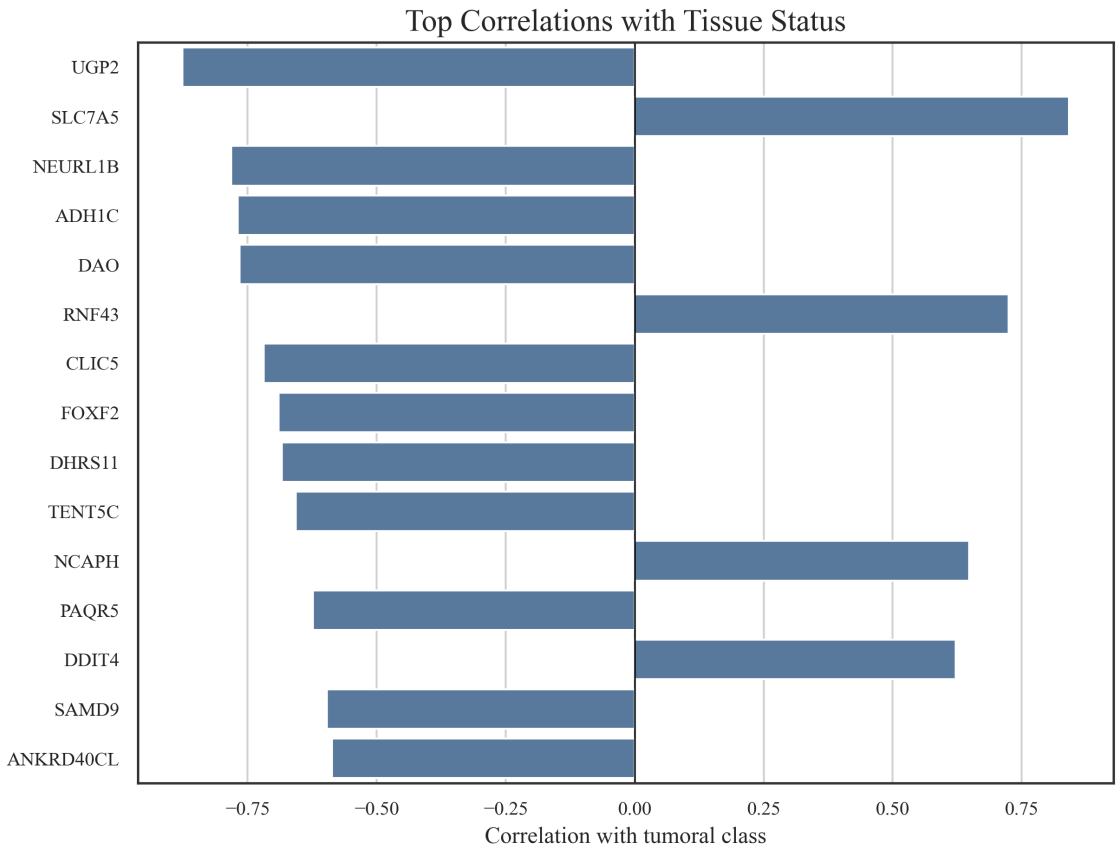


Figure 4. Ranking of the features most correlated with the target.

I used this graph to identify which variables are linearly associated with the tumoral class. It does not replace the model, but it gives a strong early signal about which genes are likely to matter. It also made the later coefficient ranking easier to interpret.

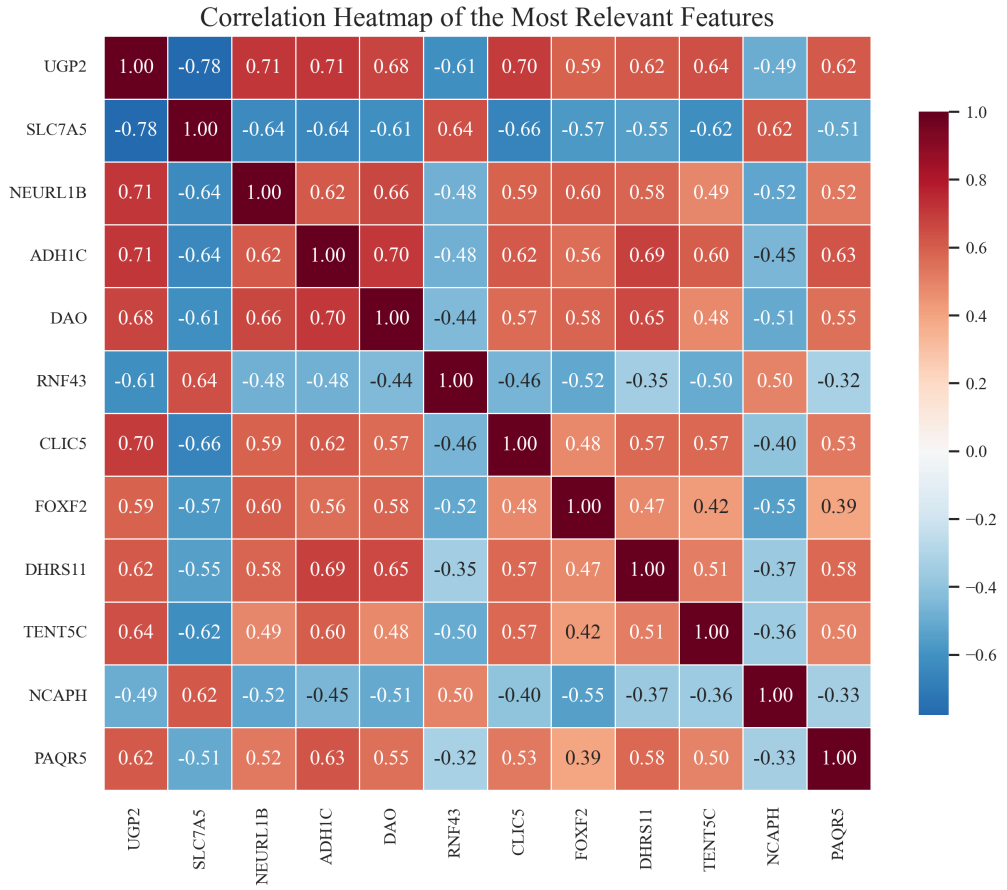


Figure 5. Heatmap showing correlation between the strongest features.

This heatmap helped me check redundancy. Some important genes are also correlated with each other, which means they may be carrying overlapping information. That matters because a small feature subset can still perform well even when the features are not fully independent.

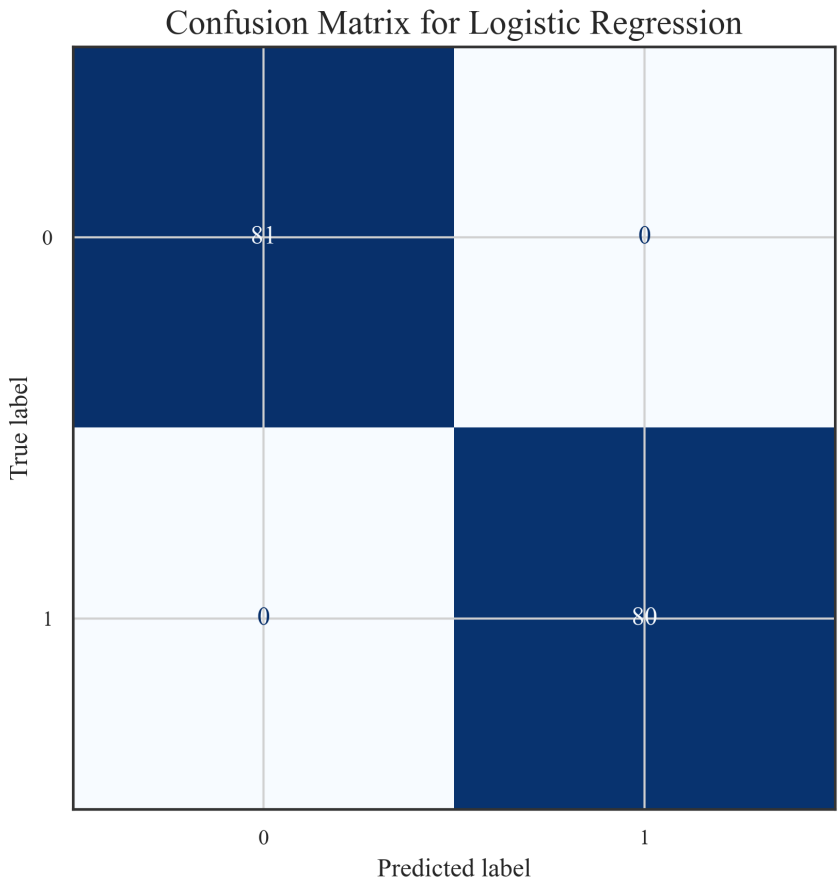


Figure 6. Confusion matrix of the full logistic-regression model.

The confusion matrix shows how many samples were correctly or incorrectly classified. In this split, the model classified the test set perfectly. That tells me the dataset contains a very strong predictive signal, at least under the current train-test partition.

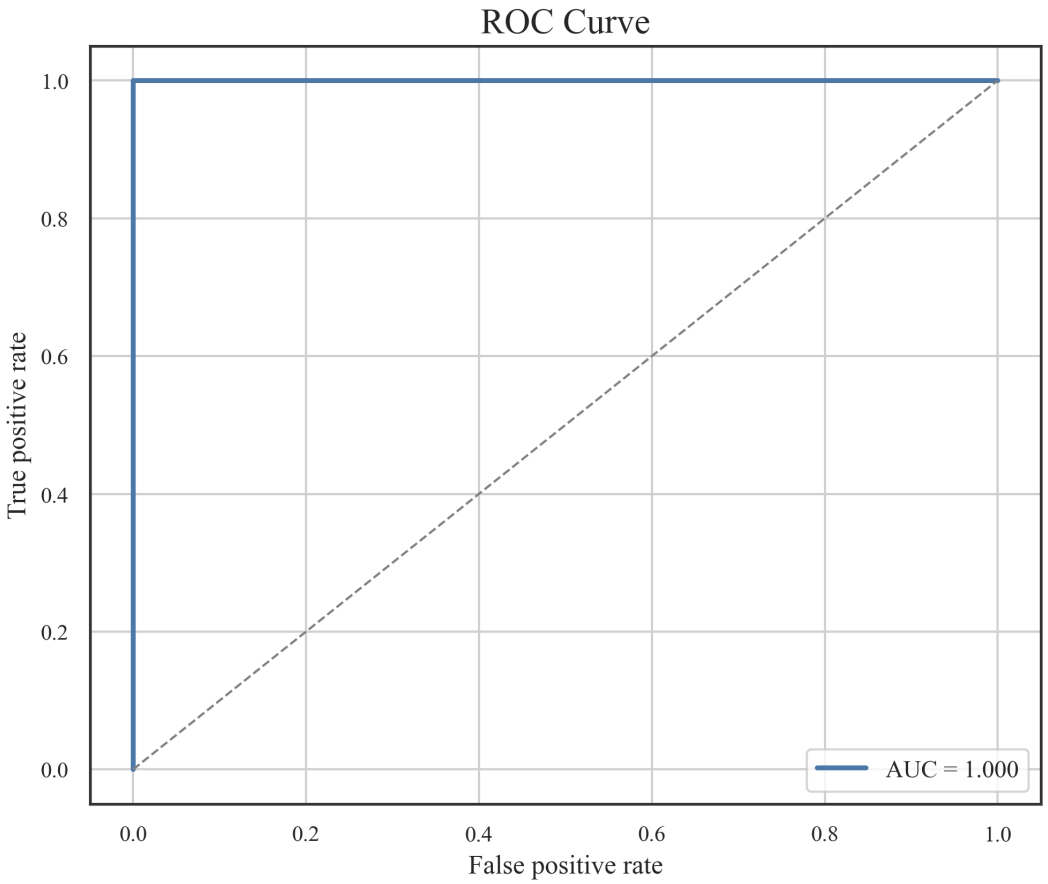


Figure 7. ROC curve for the full logistic-regression model.

The ROC curve confirms the strength of the classifier across many thresholds instead of only one. Since the AUC is extremely high, I can say the model is not only predicting the right labels, but also ranking the observations very well by class probability.

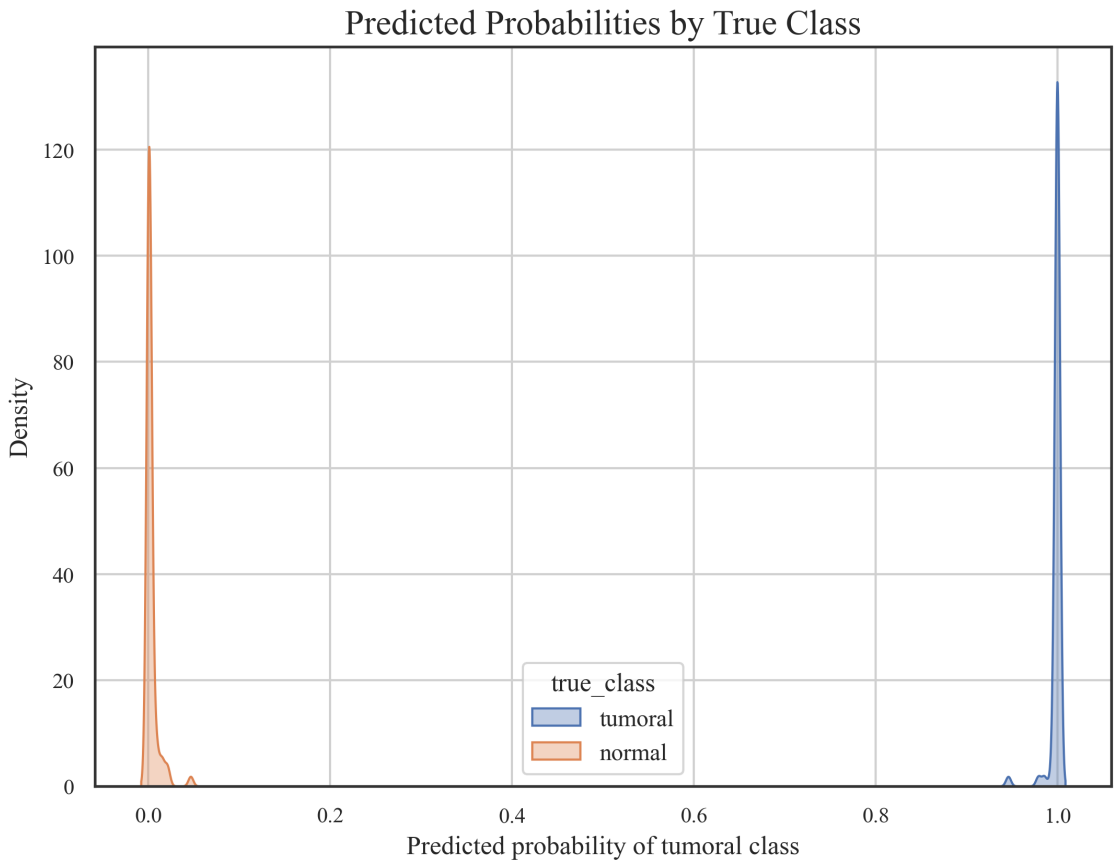


Figure 8. Density of predicted probabilities for each true class.

I included this graph because it gives a more intuitive view of the model output. The two classes are concentrated in very different probability regions, which means the good classification result is not coming from borderline cases. The model is actually very confident on most observations.

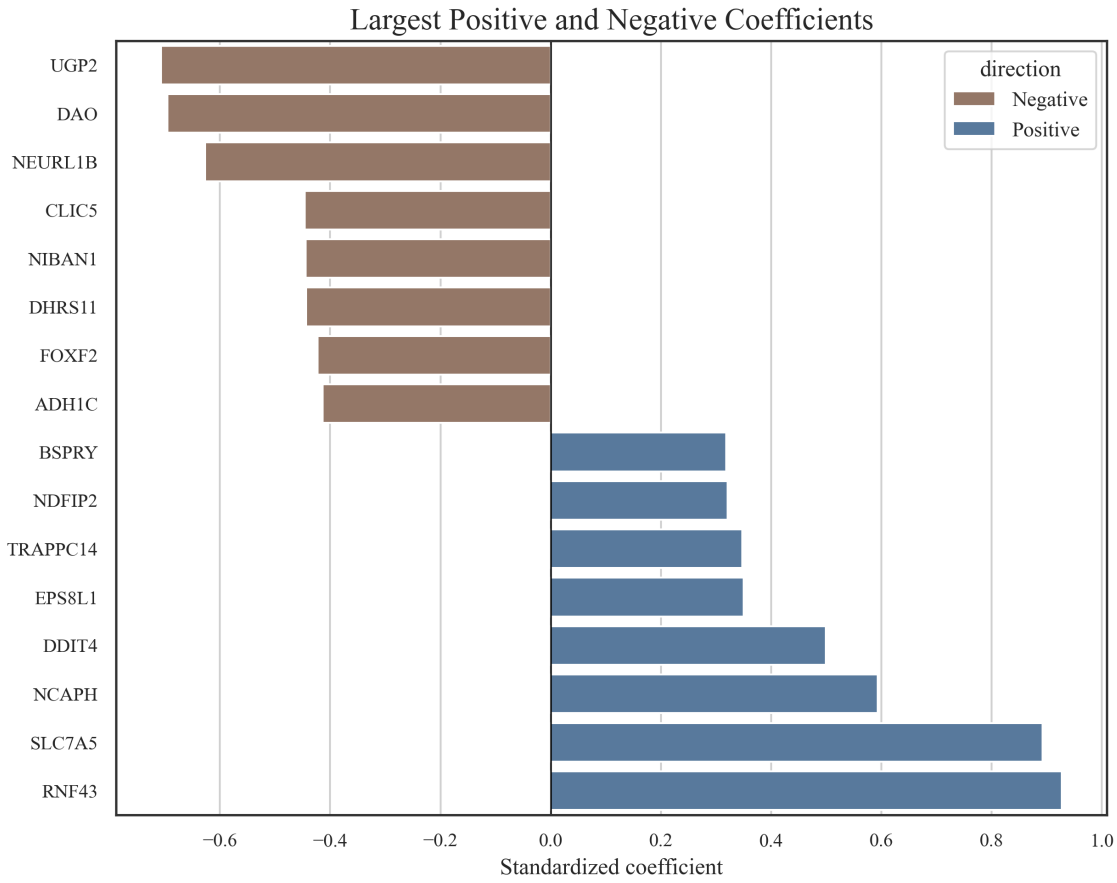


Figure 9. Signed coefficient plot for the logistic-regression model.

This graph shows direction as well as magnitude. Positive coefficients support the tumoral class, while negative ones push predictions toward the normal class. I used it to understand not only which genes matter, but also in what direction they influence the model.

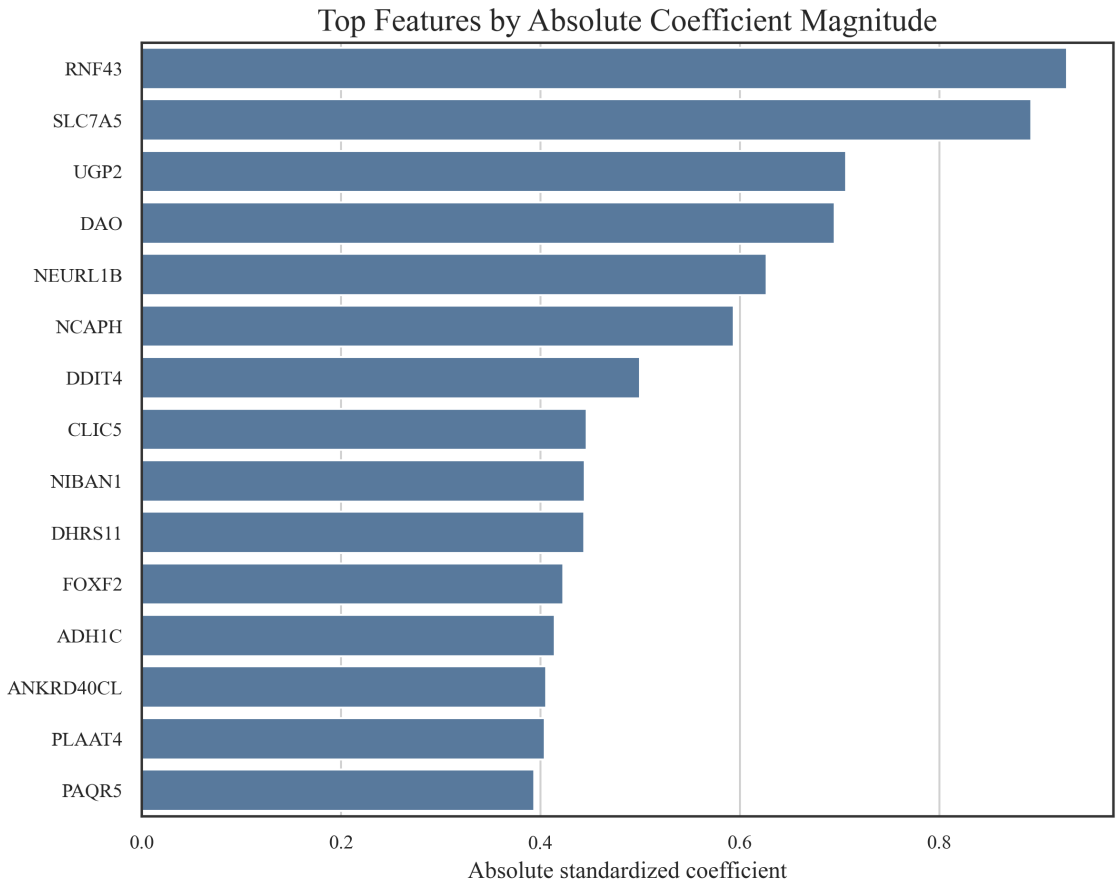


Figure 10. Ranking of the top features by absolute coefficient value.

This was the main ranking figure for feature selection. I used absolute coefficient size because I wanted to keep the strongest variables regardless of direction. It gave me a simple and consistent way to build the top 3, top 5, and top 10 reduced models.

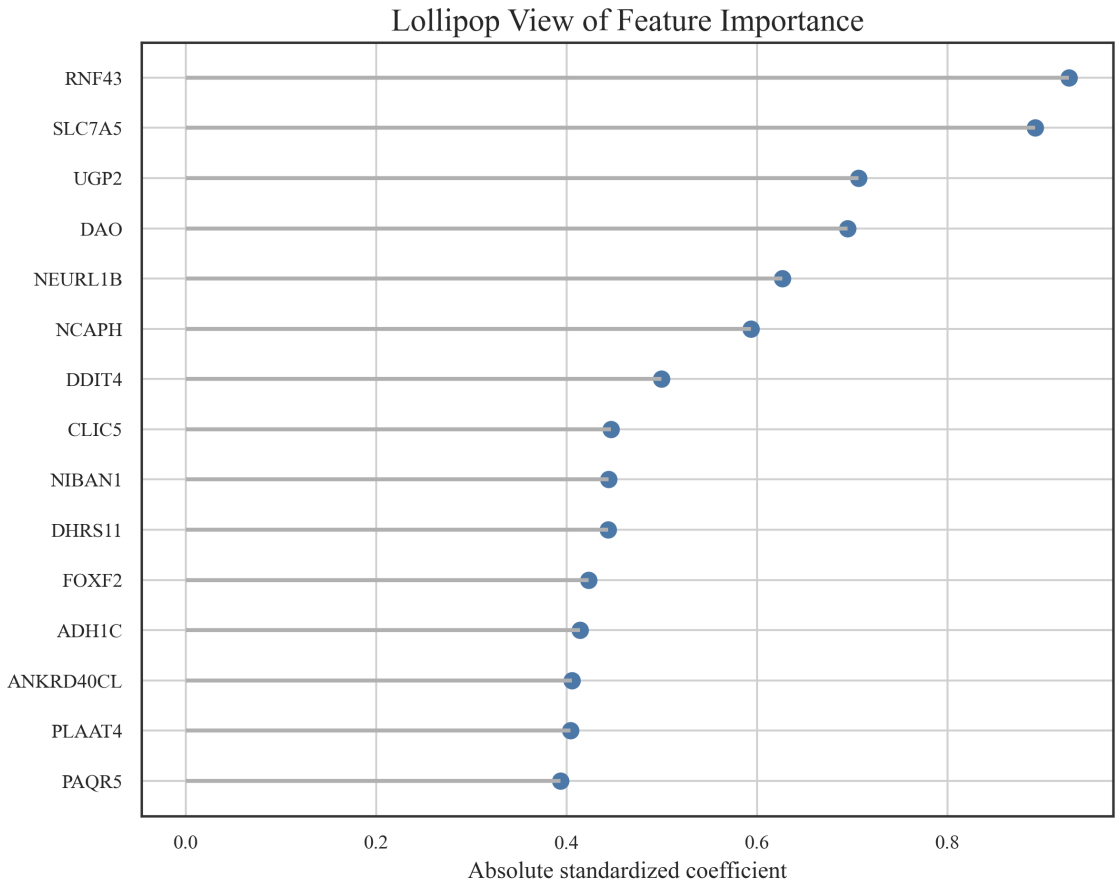


Figure 11. Lollipop representation of feature importance.

I kept this version because it is visually cleaner than a dense bar chart. The spacing between the points makes it easier to compare the top variables and to notice where there are large or small gaps in importance from one feature to the next.

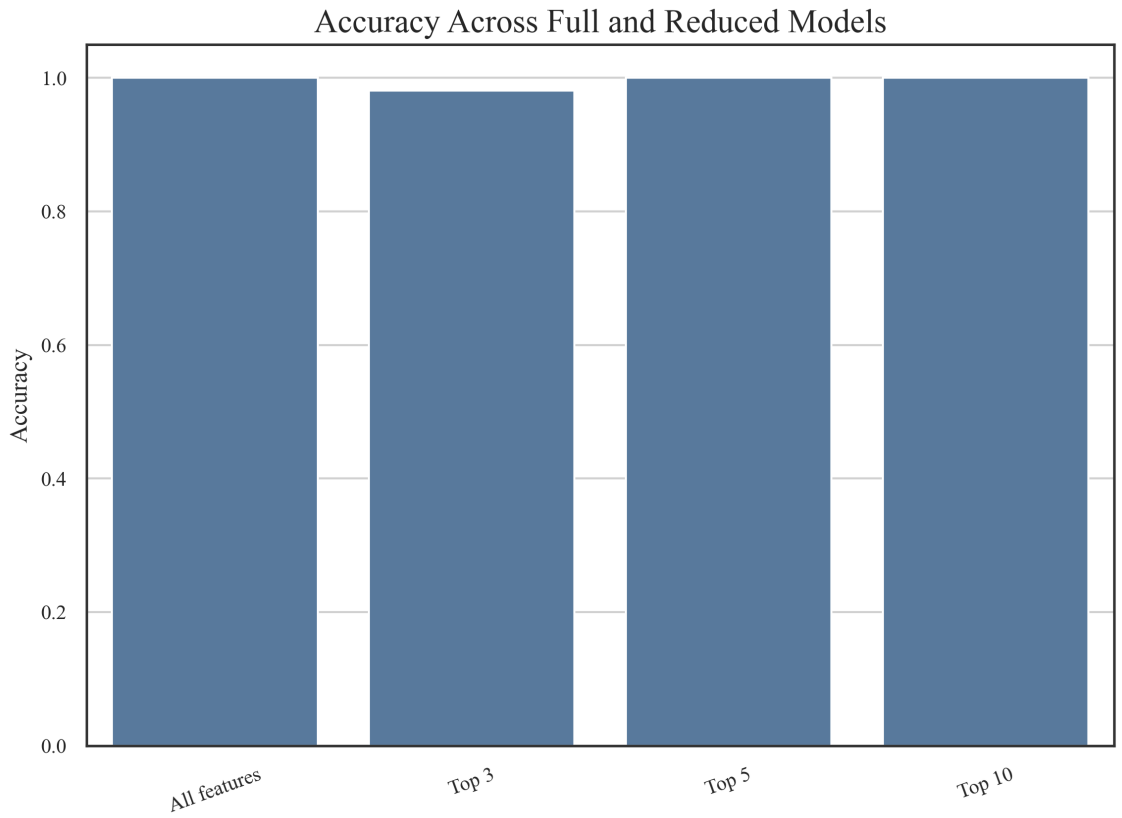


Figure 12. Accuracy comparison between the full and reduced models.

This figure shows that reducing the number of features does not necessarily destroy performance. The reduced models stayed very strong, especially the top 5 and top 10 versions. That is one of the most important findings from the first lab.

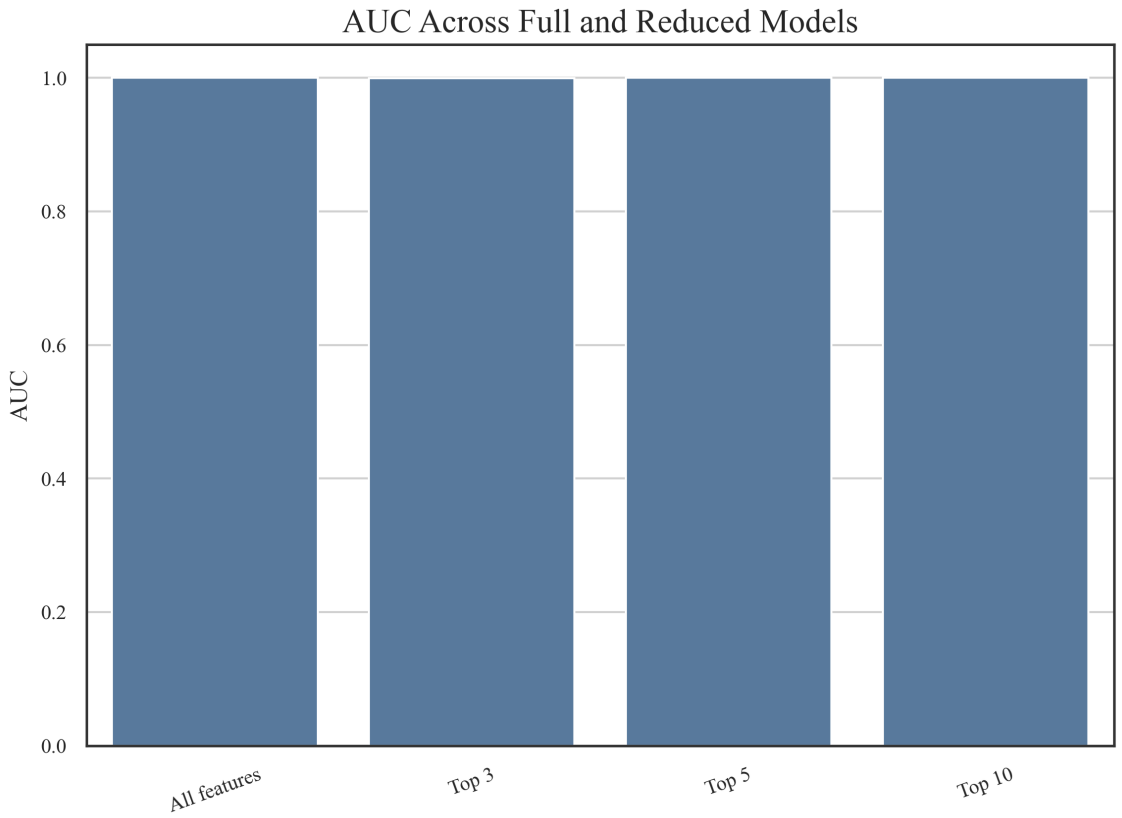


Figure 13. AUC comparison between the full and reduced models.

I used AUC together with accuracy because it is more informative about class separation. Even after reducing the feature space, the AUC remained extremely high, which supports the idea that most of the useful signal is concentrated in a small subset of genes.

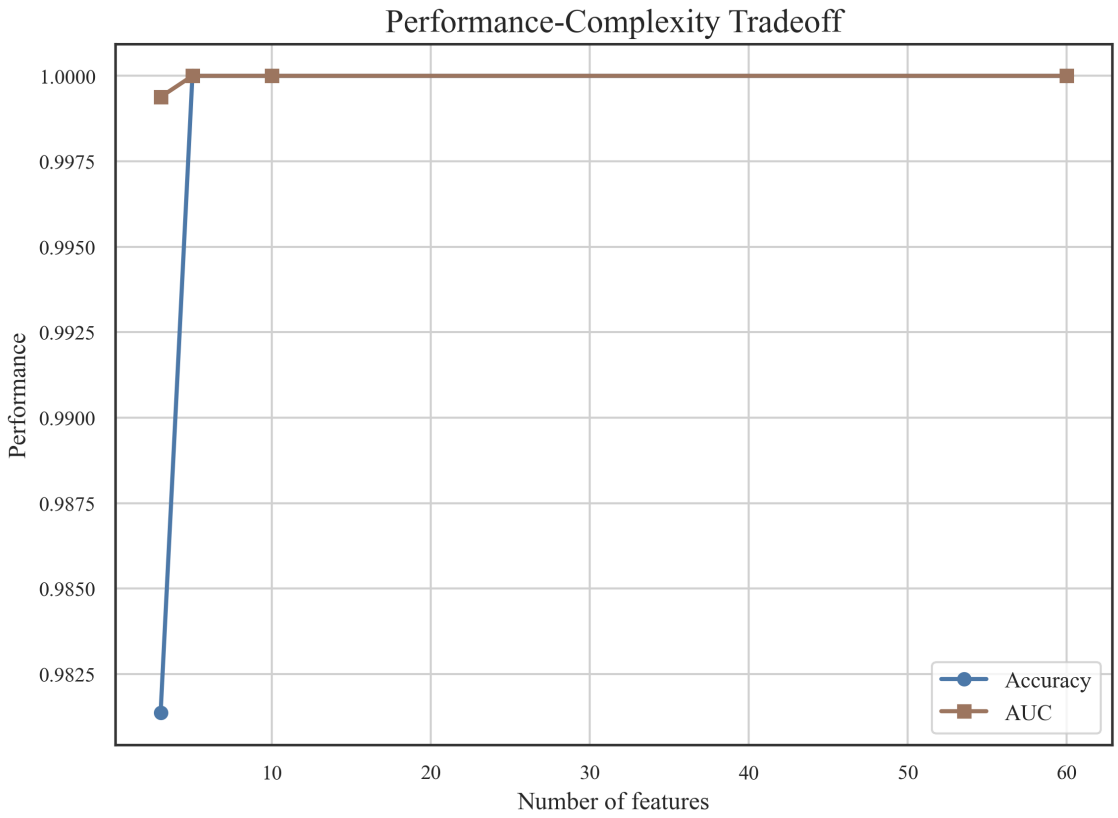


Figure 14. Tradeoff between performance and number of selected features.

This graph summarizes the practical side of feature selection. A smaller model is easier to explain and easier to manage, but I still want strong predictive quality. Here the curves stay high, which means the reduction is efficient and not too costly.

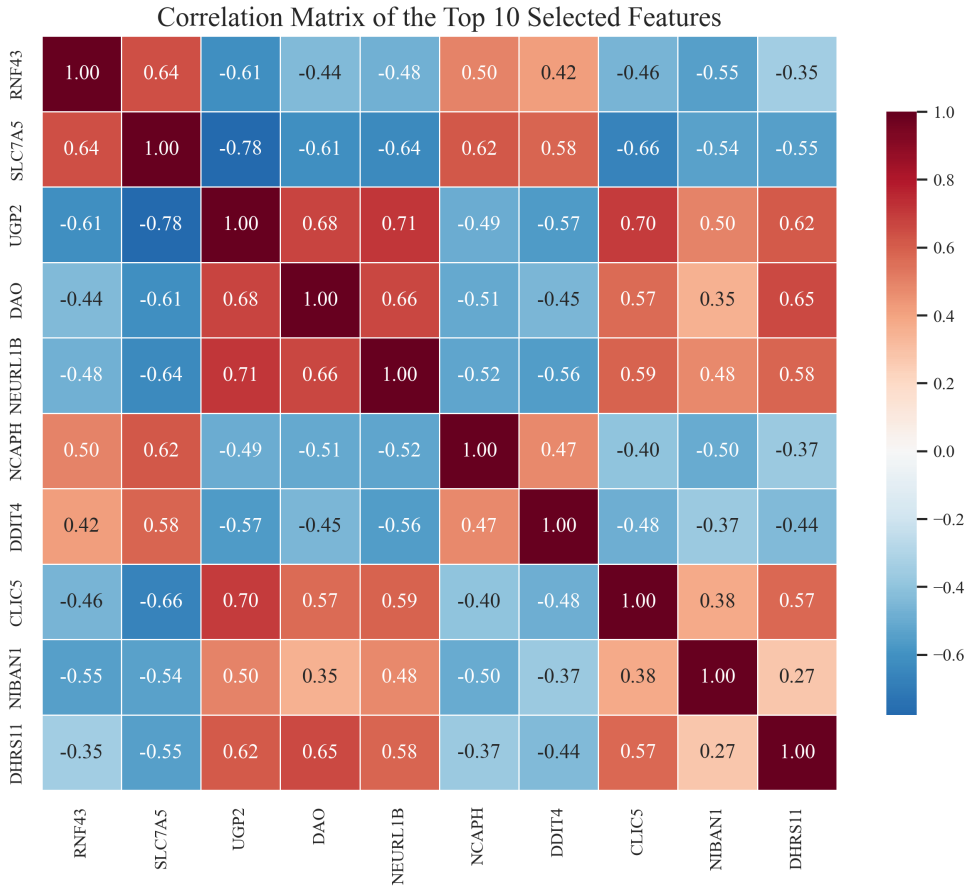


Figure 15. Correlation matrix of the selected top 10 genes.

I ended the first lab with this heatmap because it shows the dependency structure inside the reduced feature set. Some genes still move together, so even a small model can contain redundancy. This is important to mention when discussing biological interpretation.

Lab 2 - PCA and Dimensionality Reduction

In Lab 2, I used PCA in two different ways. For the decathlon dataset, PCA helped me interpret latent performance structure between event variables and between athletes. For the digits dataset, PCA was used as a reduction step before classification so that I could compare model performance before and after compression.

Lab 2 Method Summary

The second lab contains both an exploratory method and a predictive comparison. I therefore added one table for the decathlon PCA summary and another for the digits classification comparison.

Method	Measure	Value	Reference
PCA on decathlon	PC1	0.327	0.327
PCA on decathlon	PC2	0.174	0.501
PCA on decathlon	Components for 80% variance	5	0.800
PCA on decathlon	Components for 90% variance	7	0.900

Part A - Decathlon PCA

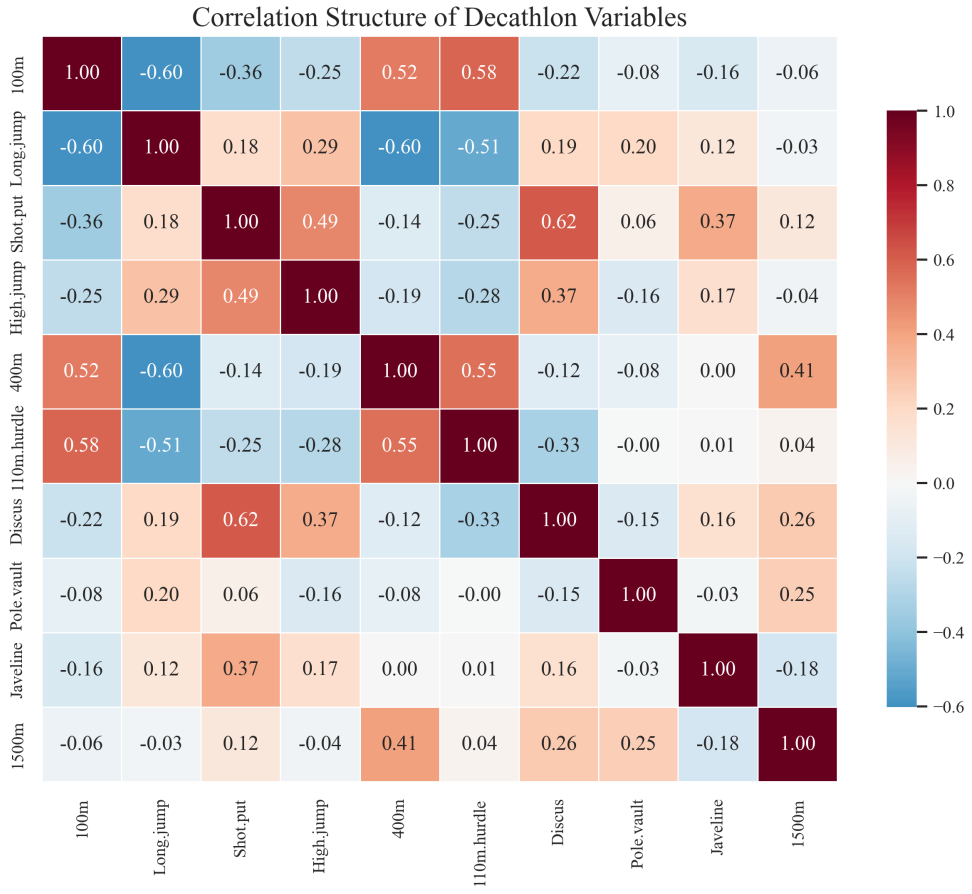


Figure 16. Correlation heatmap of the decathlon event variables.

I used this graph to justify PCA. Since several events are related to each other, the dataset clearly contains shared structure. PCA is useful in this situation because it summarizes that structure into a smaller set of components.

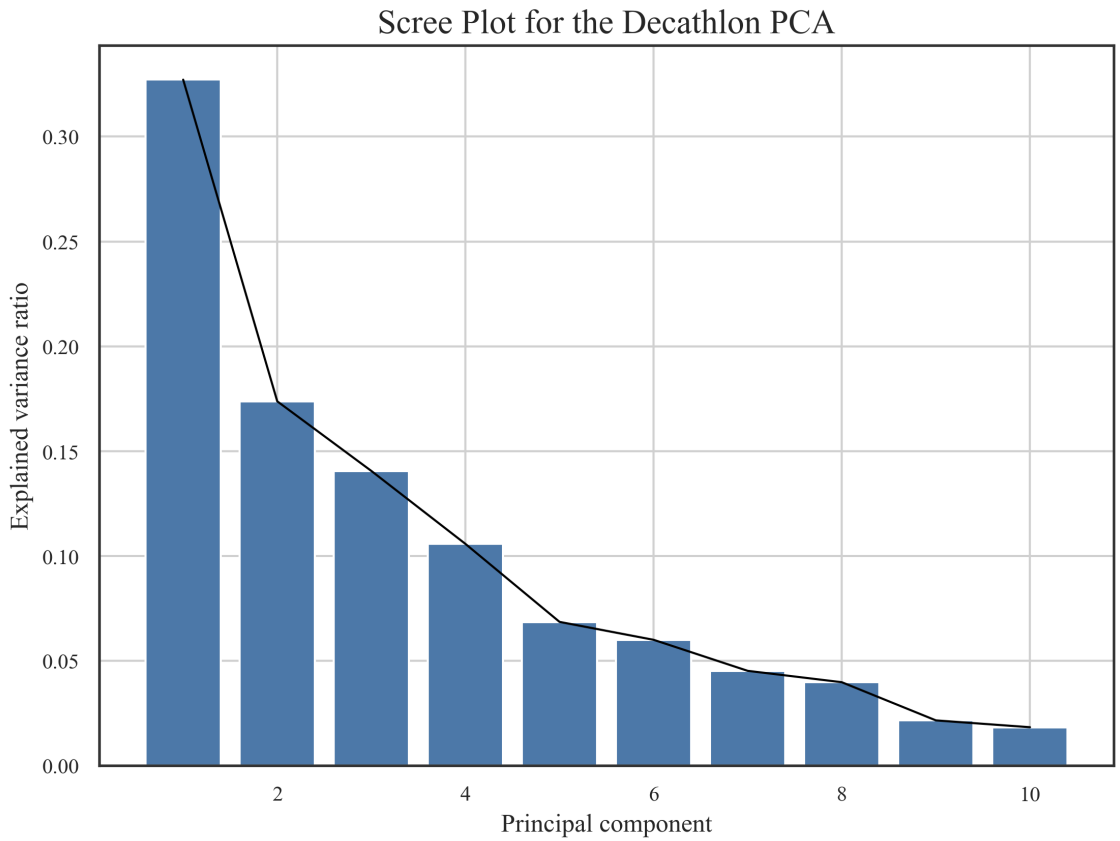


Figure 17. Scree plot of the decathlon PCA.

This scree plot shows how much variance is explained by each component separately. The first component is the strongest, but the drop is gradual, which means the full structure of the athletes cannot be reduced to just one dimension.

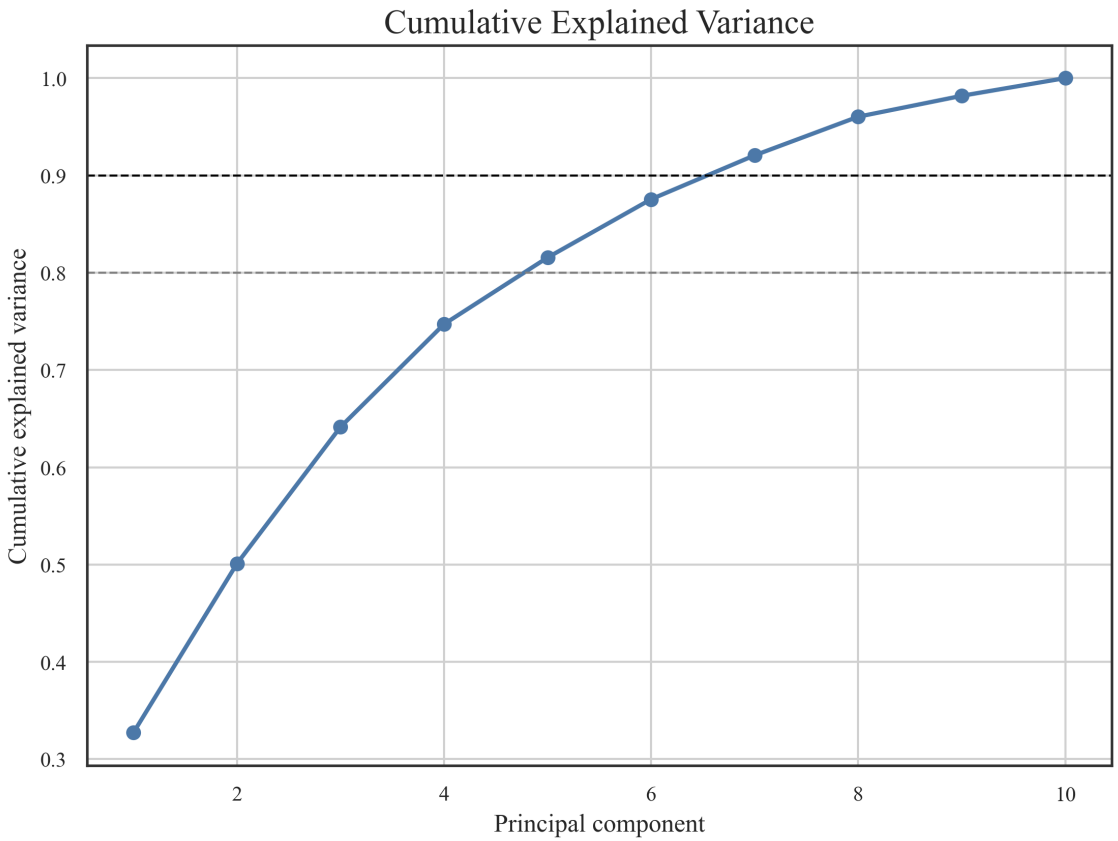


Figure 18. Cumulative explained variance for the decathlon PCA.

I used this figure to decide how many components are enough depending on the goal. Two components are useful for visualization, but more are needed if I want to keep most of the total variance. This is why interpretation and compression are not always the same decision.

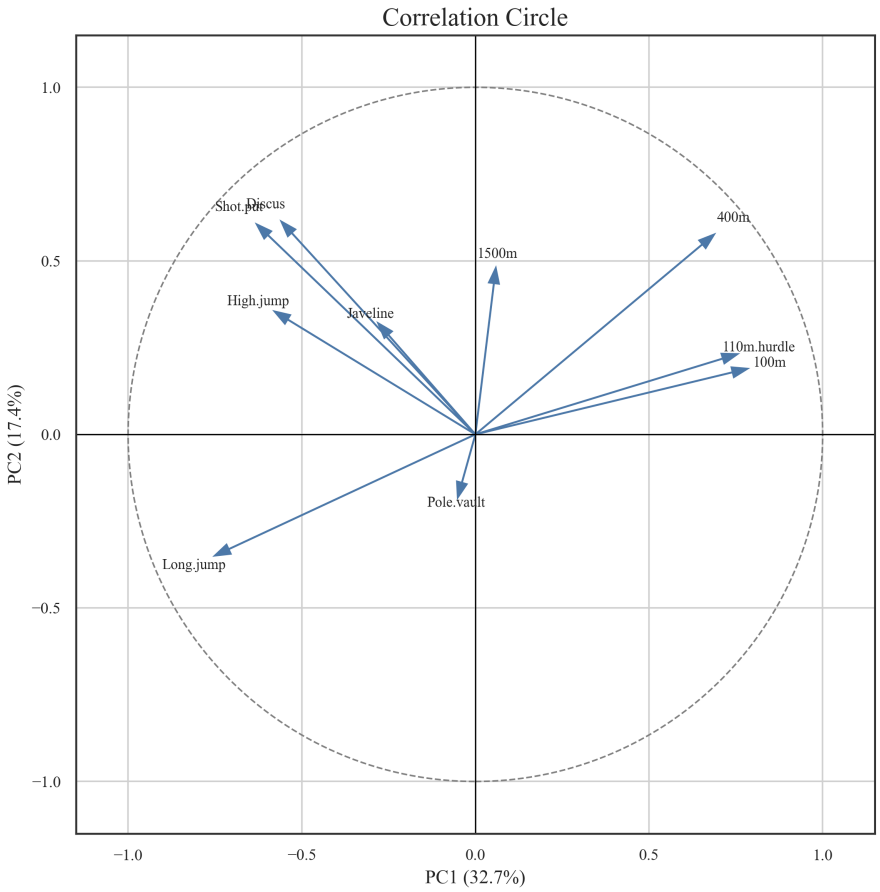


Figure 19. Correlation circle of the first two principal components.

This is one of the most informative PCA graphs. It shows how the original event variables behave in the new component space. I noticed that timed events and distance-based events often point in opposite directions, which makes sense because good performance is measured differently in those groups.

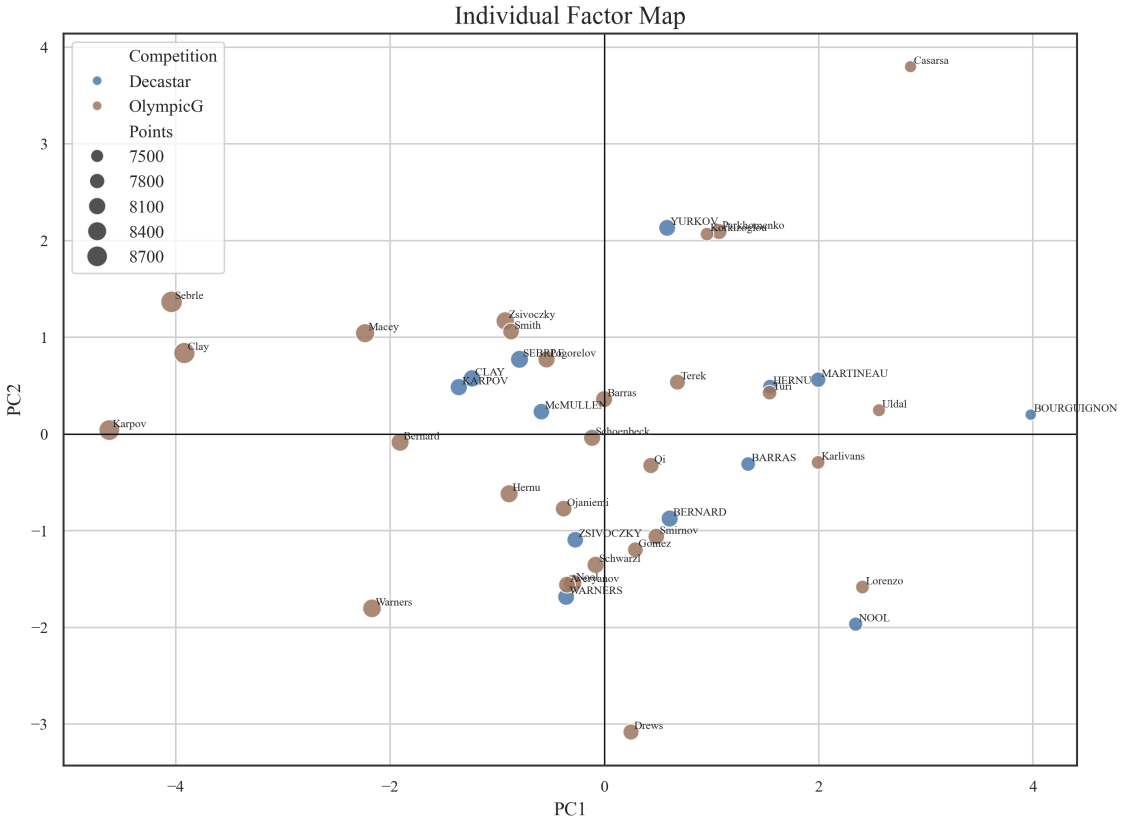


Figure 20. Factor map of the athletes on the first two principal components.

This figure allowed me to compare the athletes directly. Athletes that are close on the map have similar standardized performance profiles, while athletes that are far apart have more different balances across the events.

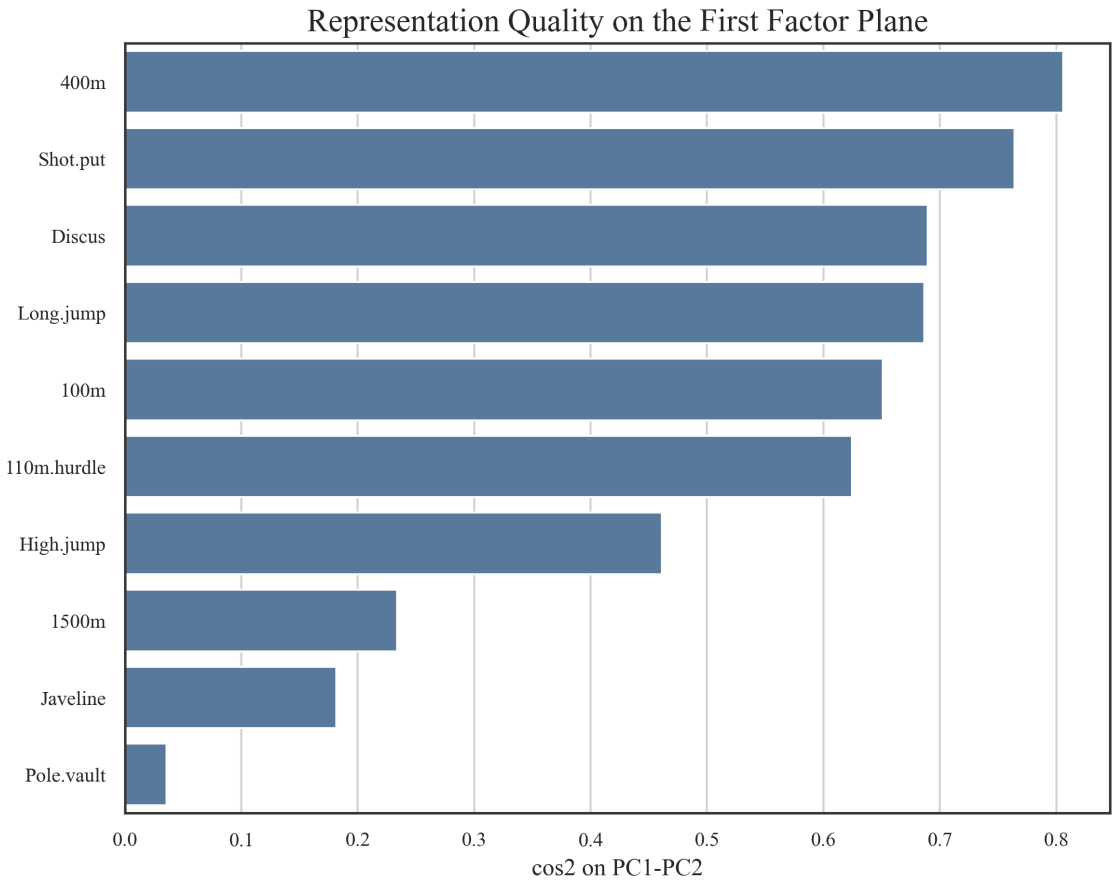


Figure 21. Quality of representation of the variables on PC1 and PC2.

I used cos2 values to check which variables are well represented on the first factor plane. This is important because not every variable is equally well summarized by the first two components. A strong PCA graph should always be read together with this kind of quality check.

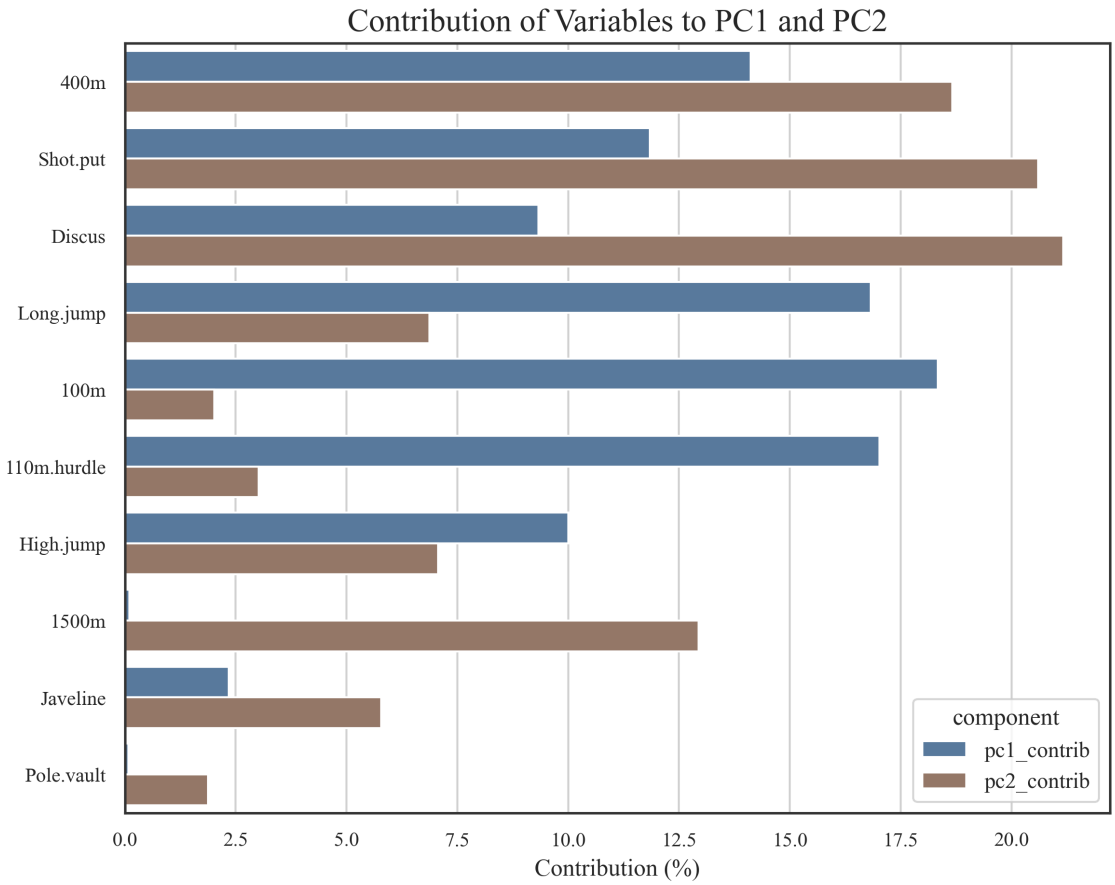


Figure 22. Variable contributions to the first two components.

This graph complements the cos2 view. Instead of asking which variables are well represented, it asks which variables are really driving the components. That helped me describe PC1 and PC2 in a more concrete way.

Part B - Digits PCA and Classification

Model	Dimensions	Accuracy	Retained variance
Without PCA	64	0.964	1.000
With PCA	31	0.962	0.903

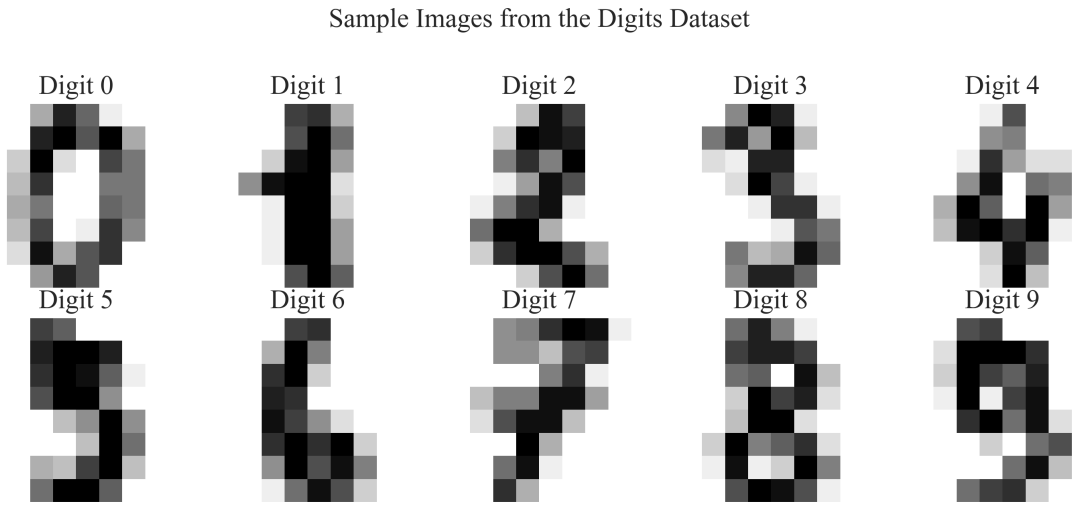


Figure 23. Example images from the digits dataset.

I added the raw images because they remind me that this dataset is still a visual dataset even if the model sees a vector of numbers. Looking at the original digits makes the PCA results easier to connect to something concrete.

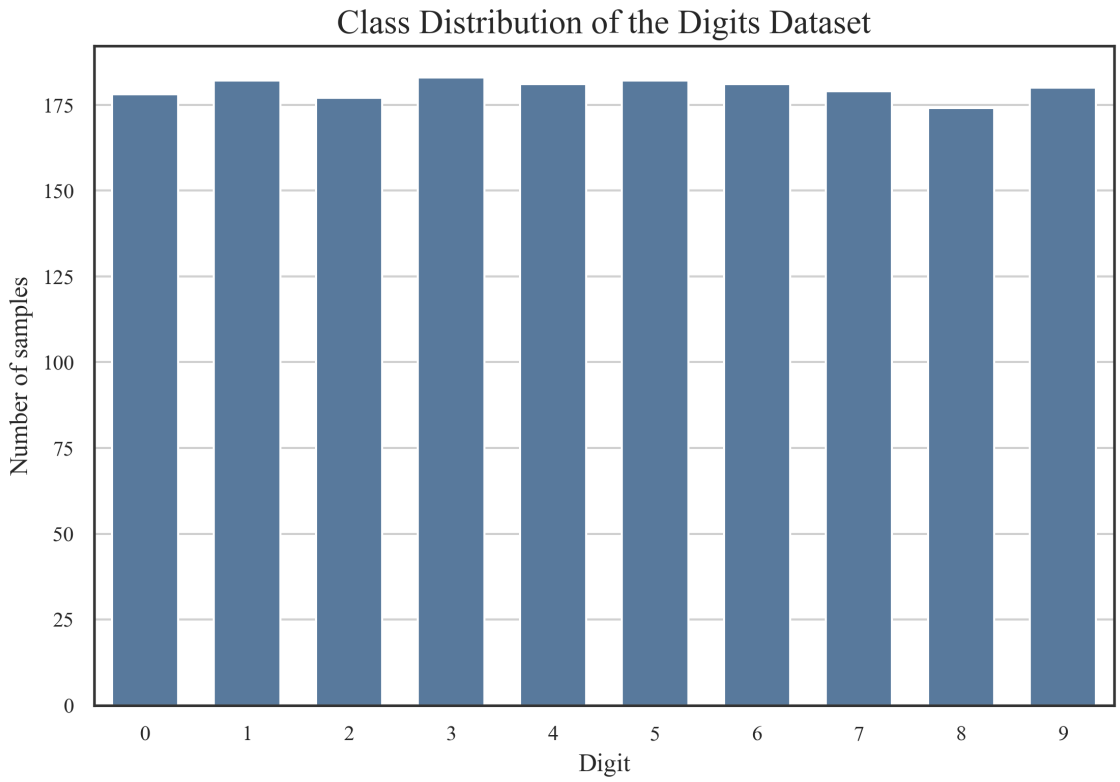


Figure 24. Distribution of the ten digit classes.

The digits dataset is also well balanced, which is useful for the classification comparison. This means that later performance differences are not being distorted by one class having much more data than the others.

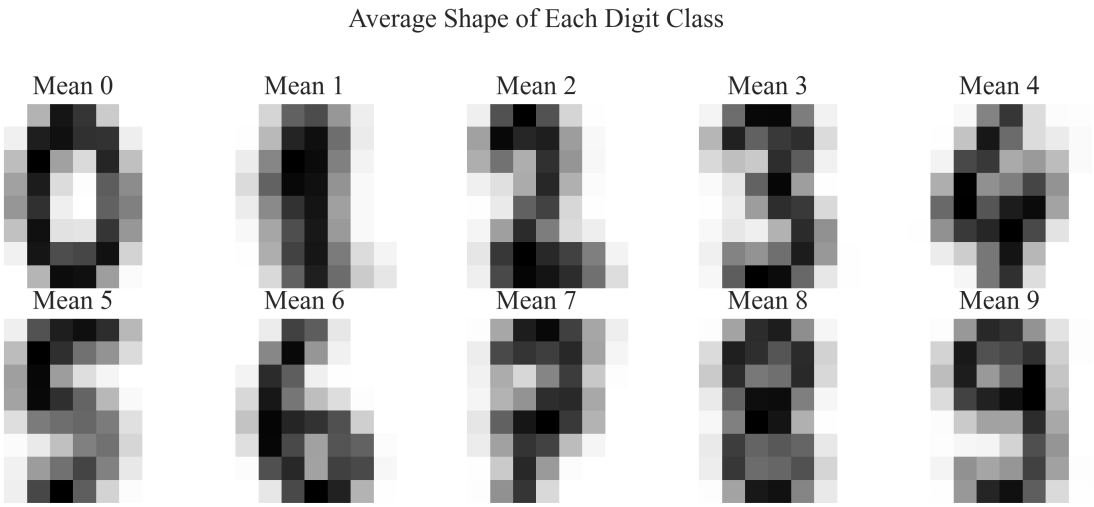


Figure 25. Average image for each class in the digits dataset.

This figure helped me understand the stable shape patterns inside each class. Some digits are visually very distinct, while others are closer to each other. That makes it easier to understand why some confusions are more likely than others.

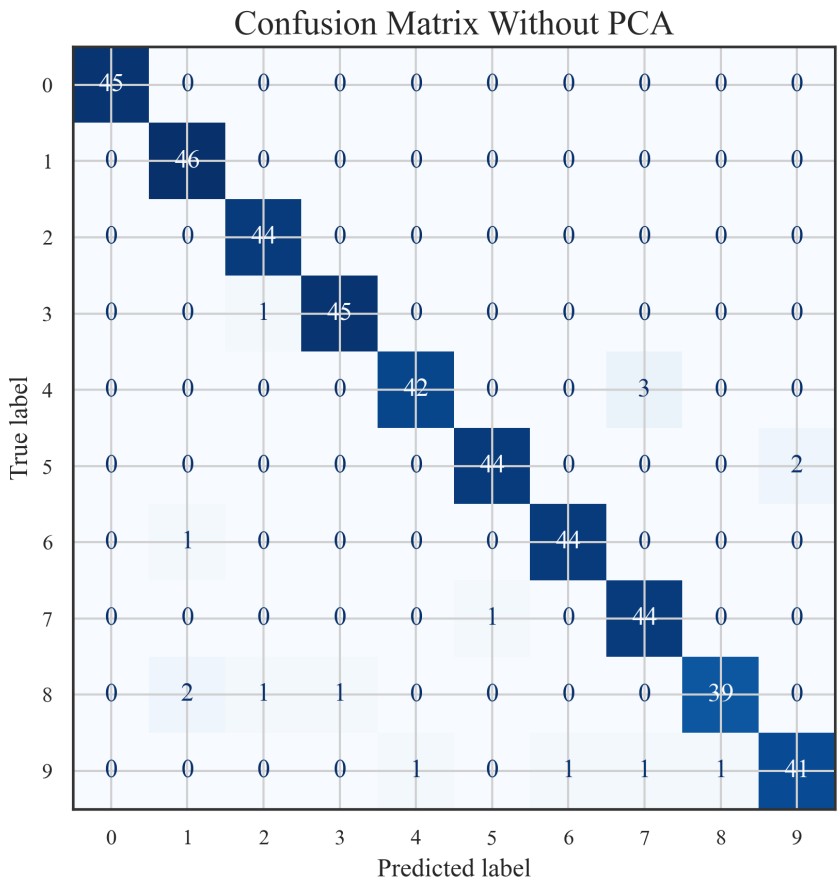


Figure 26. Confusion matrix for k-NN before PCA.

This is the baseline result for the digits task. I used it as the reference point before applying PCA. It shows which digit classes are easiest and hardest for the classifier when it uses the original 64-dimensional input space.

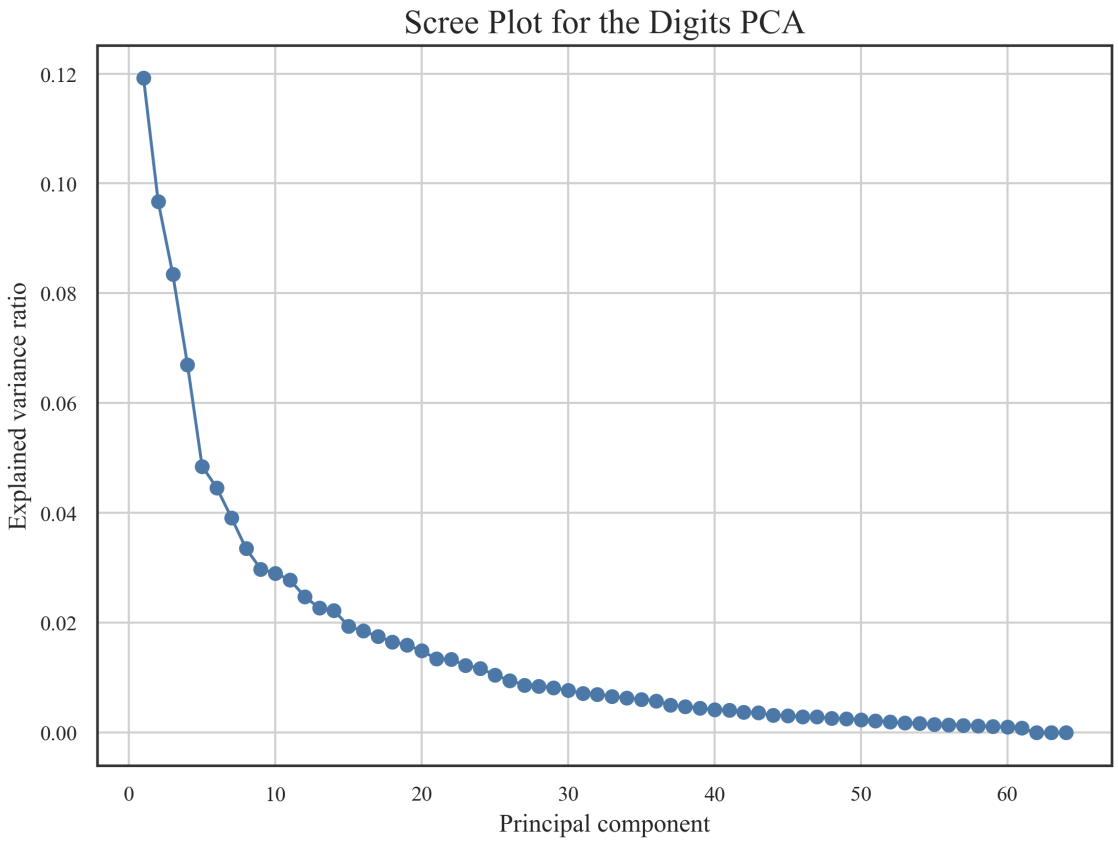


Figure 27. Scree plot for the digits PCA.

This scree plot shows how much information each component carries on its own. The early components are clearly the most important, but the variance is spread across several dimensions, which is common in image data.

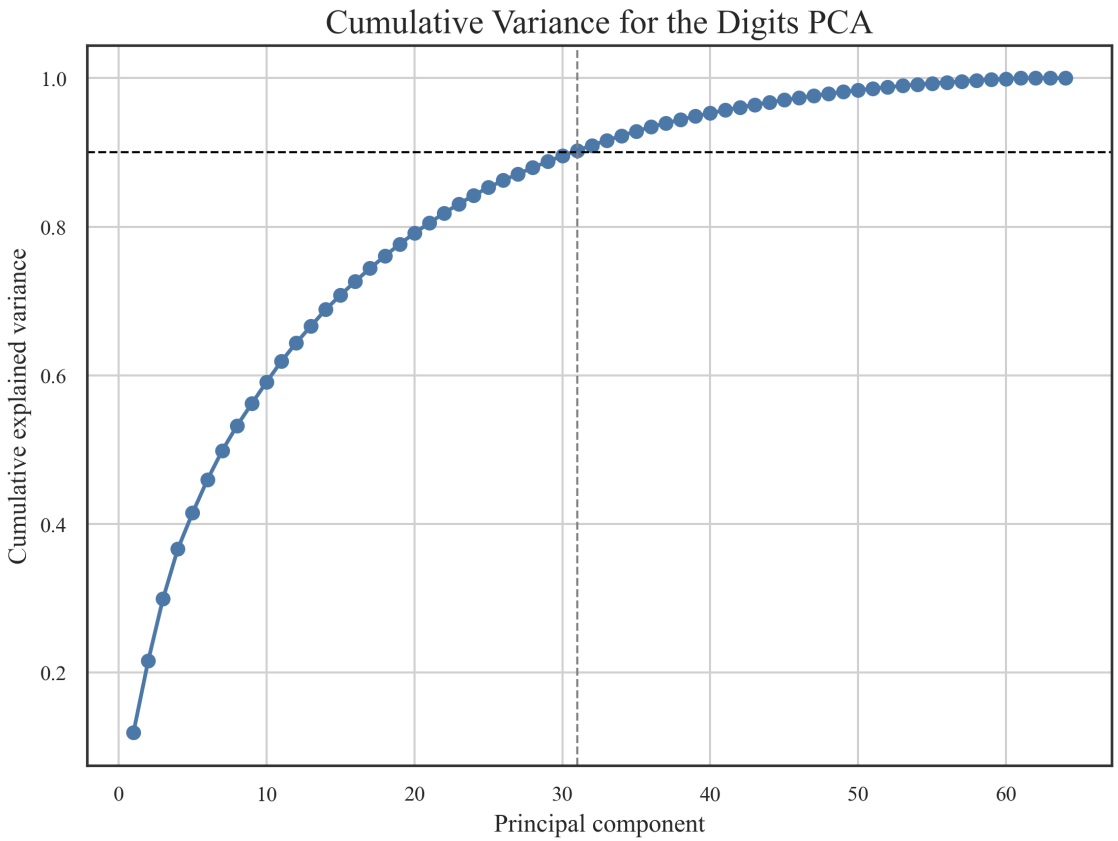


Figure 28. Cumulative explained variance of the digits PCA.

I used this curve to decide how many principal components to keep. In this run, 31 components were enough to preserve at least 90% of the variance, which means PCA reduced the dimensionality by more than half.

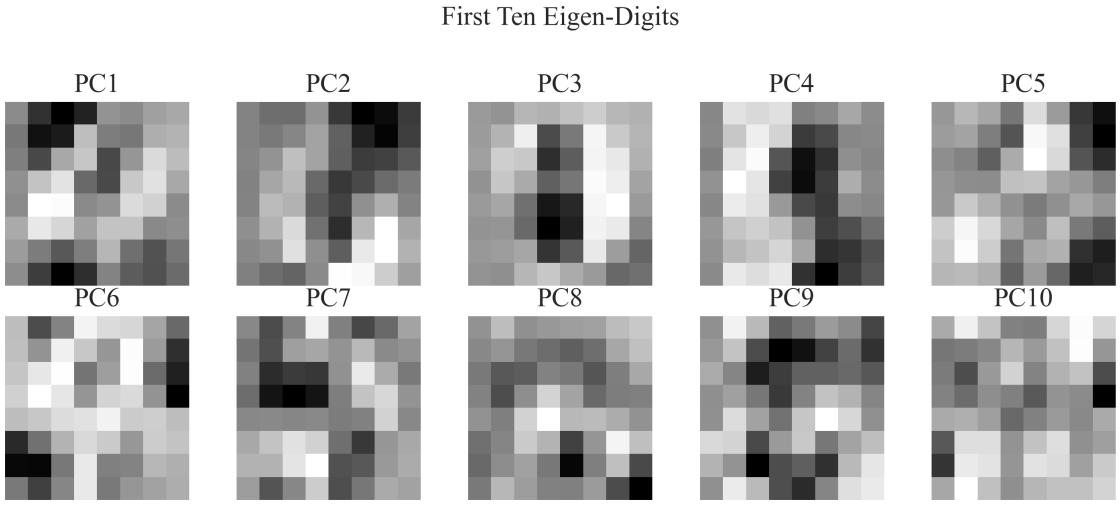


Figure 29. Visual form of the first ten principal components.

These images helped me understand PCA in a more visual way. The components are not real digits, but they show the main stroke patterns that vary across the dataset. This makes the idea of dimensionality reduction much easier to interpret.

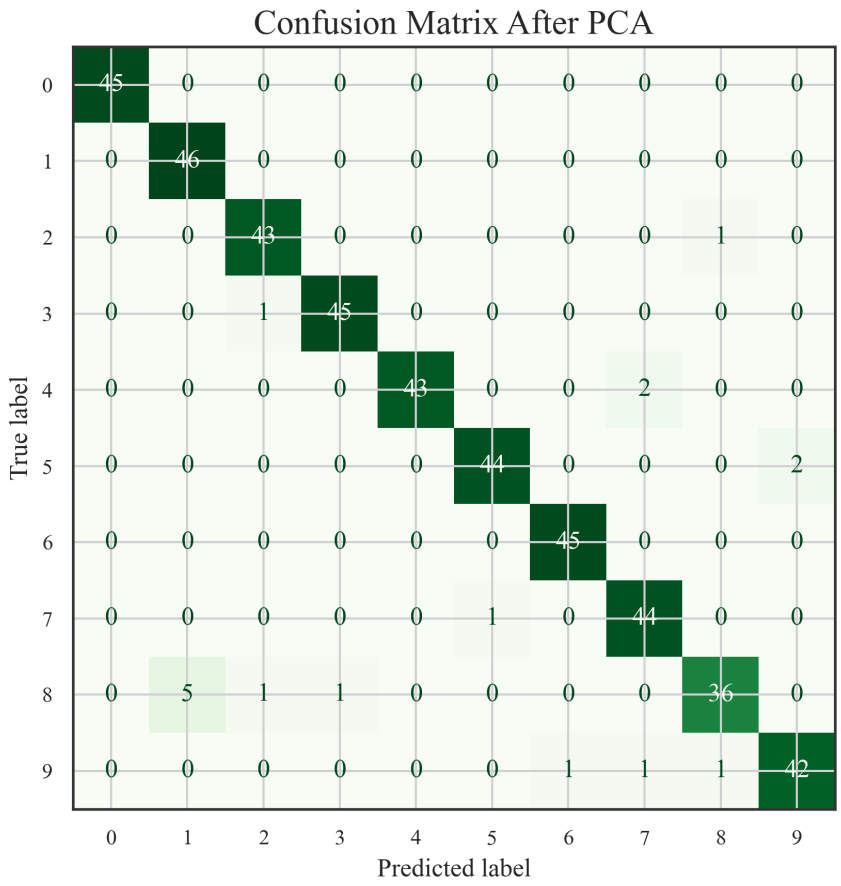


Figure 30. Confusion matrix for k-NN after PCA.

This figure shows the classification result after replacing the original features with PCA components. The matrix stays close to the baseline result, which means the projection kept most of the useful structure for prediction.

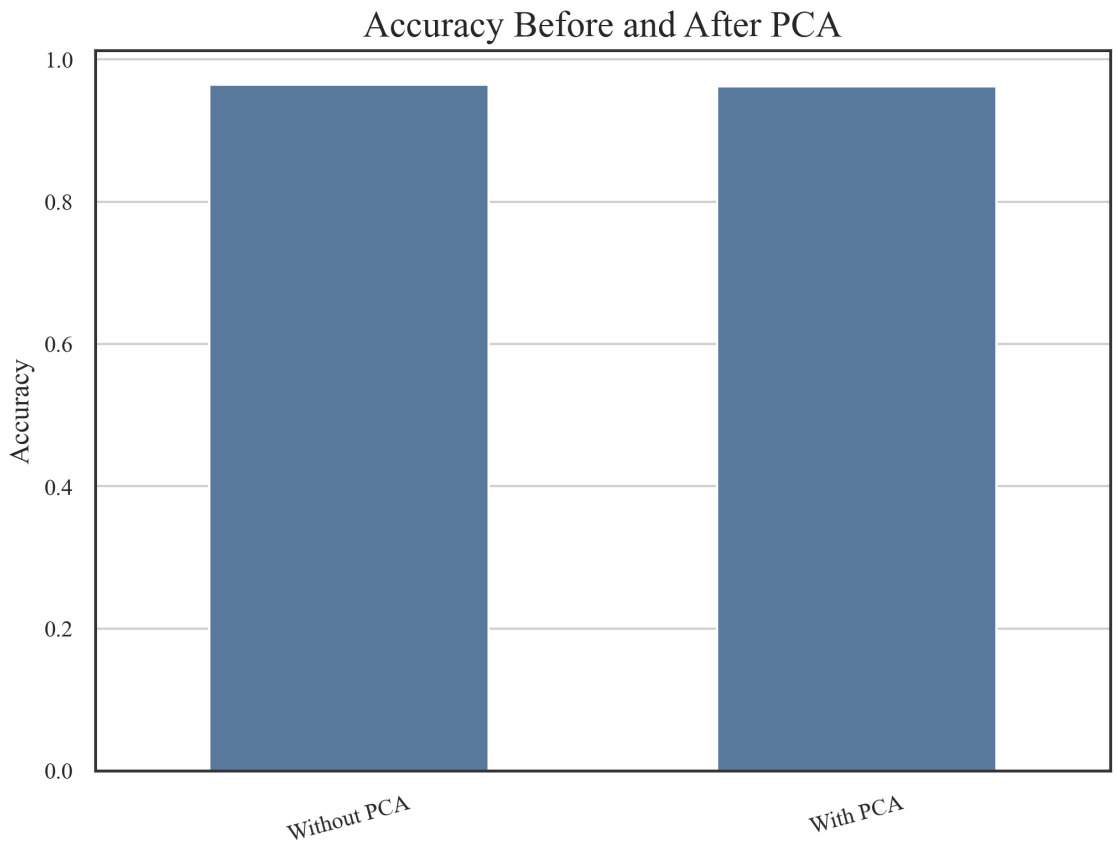


Figure 31. Accuracy comparison before and after PCA.

This graph is the clearest summary of the predictive comparison. The accuracy drops only slightly after PCA, which tells me that dimensionality reduction preserved most of the class information needed by the k-NN model.

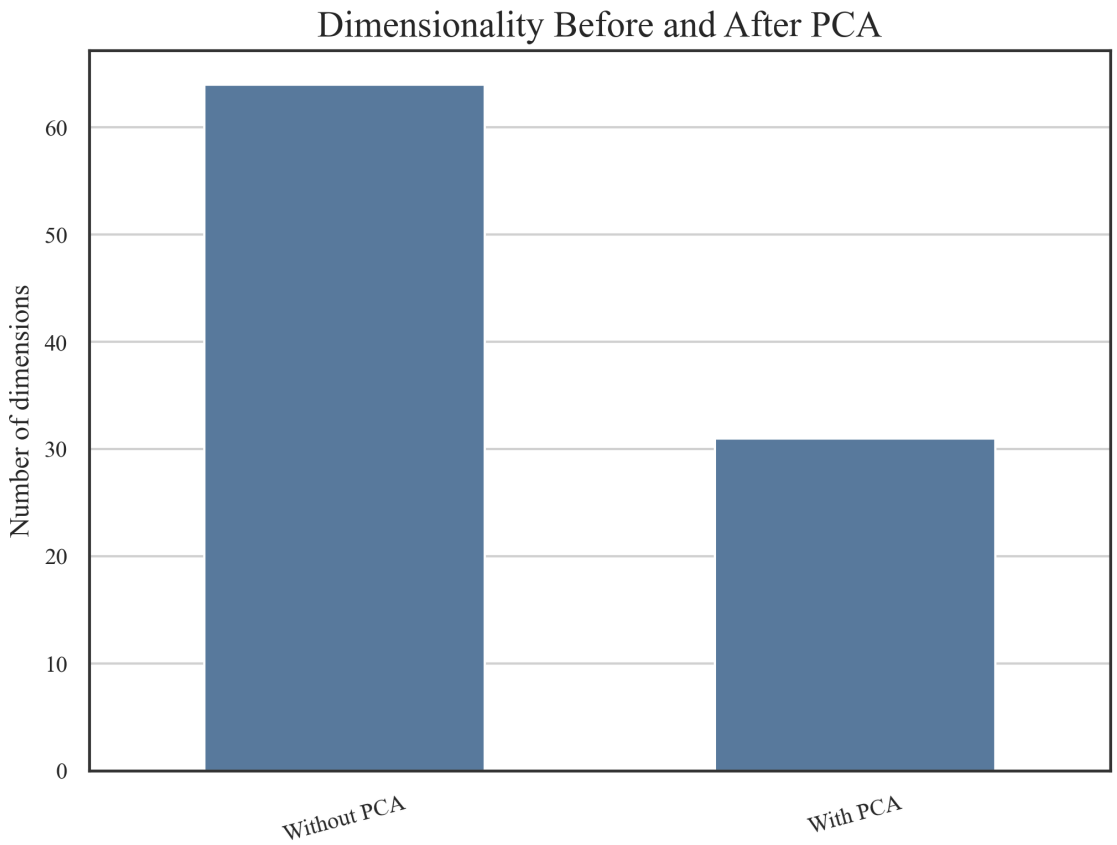


Figure 32. Dimensionality comparison before and after PCA.

This graph shows the size of the reduction very clearly. The dataset goes from 64 original variables to 31 principal components while keeping around 90% of the variance. This is the main compression benefit of PCA in the digits problem.

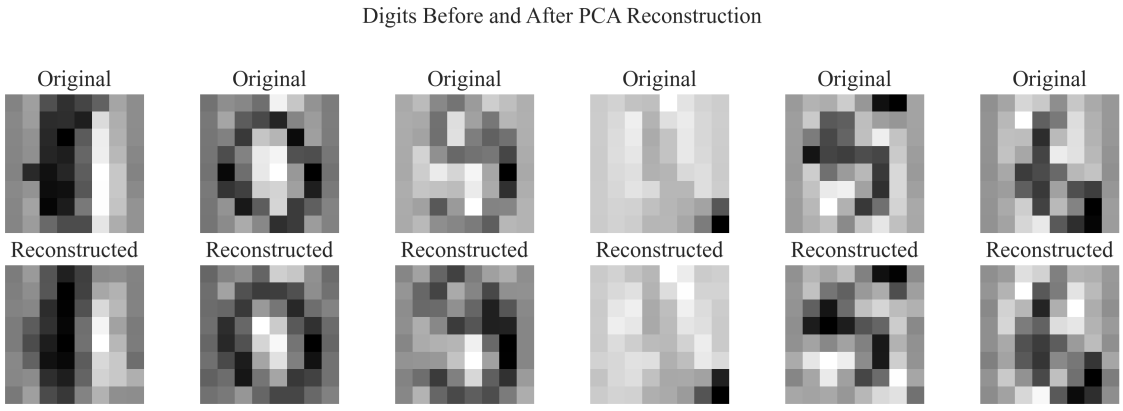


Figure 33. Reconstruction of test images after PCA compression.

I liked this figure because it shows directly what PCA removes and what it keeps. The reconstructed digits lose some detail, but the main shapes remain readable. This matches the small drop in classification accuracy.

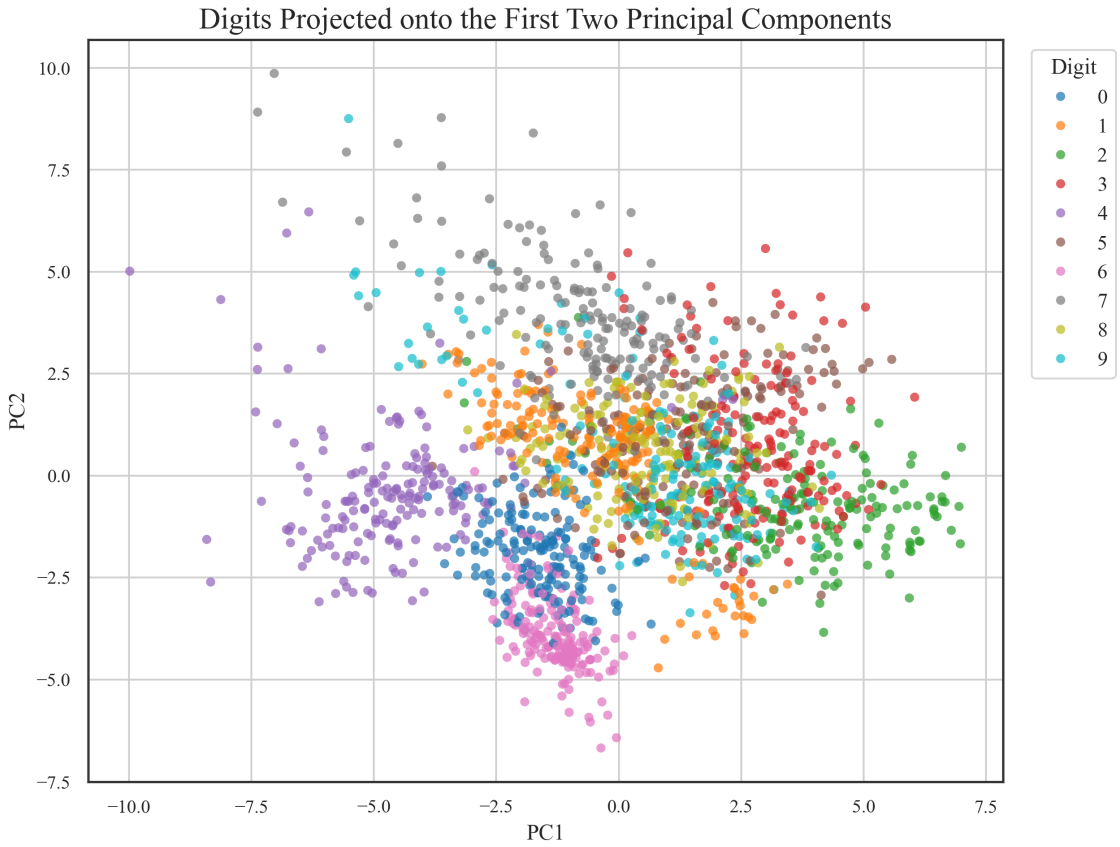


Figure 34. Two-dimensional PCA projection of the digits dataset.

This scatter plot is useful for visualization, but it also shows the limit of using only two components. Some groups are separated, but there is still overlap between several classes. So PCA in two dimensions is good for interpretation, while more dimensions are needed for stronger classification.

Cross-Lab Summary Table

Since the two notebooks do not use the same type of method, I kept the results independent but also added a compact summary table below. It brings together the main algorithms and outputs in one place so the report feels complete.

Lab	Algorithm / Method	Main output	Score or summary
Lab 1	Logistic Regression (All features)	60 features	Acc 1.000 | AUC 1.000
Lab 1	Logistic Regression (Top 3)	3 features	Acc 0.981 | AUC 0.999
Lab 1	Logistic Regression (Top 5)	5 features	Acc 1.000 | AUC 1.000
Lab 1	Logistic Regression (Top 10)	10 features	Acc 1.000 | AUC 1.000
Lab 2	PCA on Decathlon	PC1 + PC2	0.327 and 0.174 explained variance
Lab 2	k-NN Without PCA	64 dims	Accuracy 0.964
Lab 2	k-NN With PCA	31 dims	Accuracy 0.962

Conclusion

Overall, the first practical session showed that logistic regression can classify the colon cancer samples very effectively and that a reduced set of genes can keep almost the same performance as the full model. The second practical session showed that PCA is useful both for interpretation and for dimensionality reduction. On the decathlon dataset, it helped summarize the main performance structure. On the digits dataset, it reduced the number of variables while keeping most of the classification quality. For me, the main takeaway from both labs is that good analysis is not only about getting a high score, but also about understanding which variables matter, how they relate to each other, and how much complexity is really necessary.